

Sheetname: CPU

CPU MC68000
CLOCK 16MHZ
DATA AND ADDRESS BUFFERS
HALT AND RESET
DTACK GLUE LOGIC
INTERRUPT

Sheetname: MEMORY

ROM 16kBytes
RAM 1MBytes

Sheetname: FOUR UARTS

TG16C554
4 UARTS.

Arquivo: fourUarts.kicad_sch

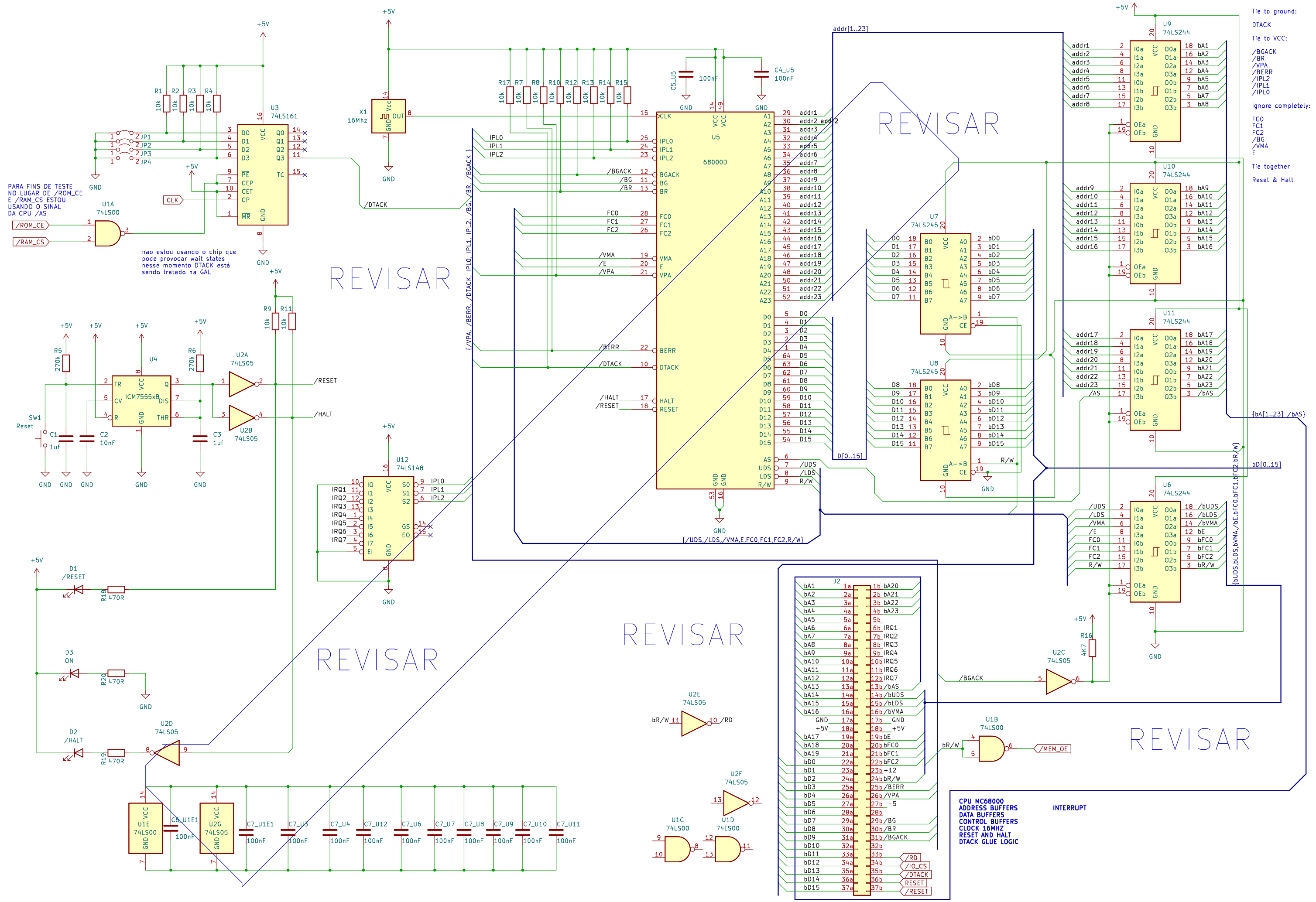
Sheetname: cpuNoBuffer

CPU MC68000
CLOCK 16MHZ
HALT AND RESET
DTACK GLUE LOGIC
INTERRUPT

Arquivo: cpuNoBuffer.kicad_sch

UART CONTROL

Arquivo: uart_control.kicad_sch



PARA FINS DE TESTE
NO LUGAR DE /ROM_CE
E /RAM_CS ESTOU
USANDO O SINAL
DA CPU /AS

nao estou usando o chip que
pode provocar wait states
nesse momento DTACK está
sendo tratado na GAL

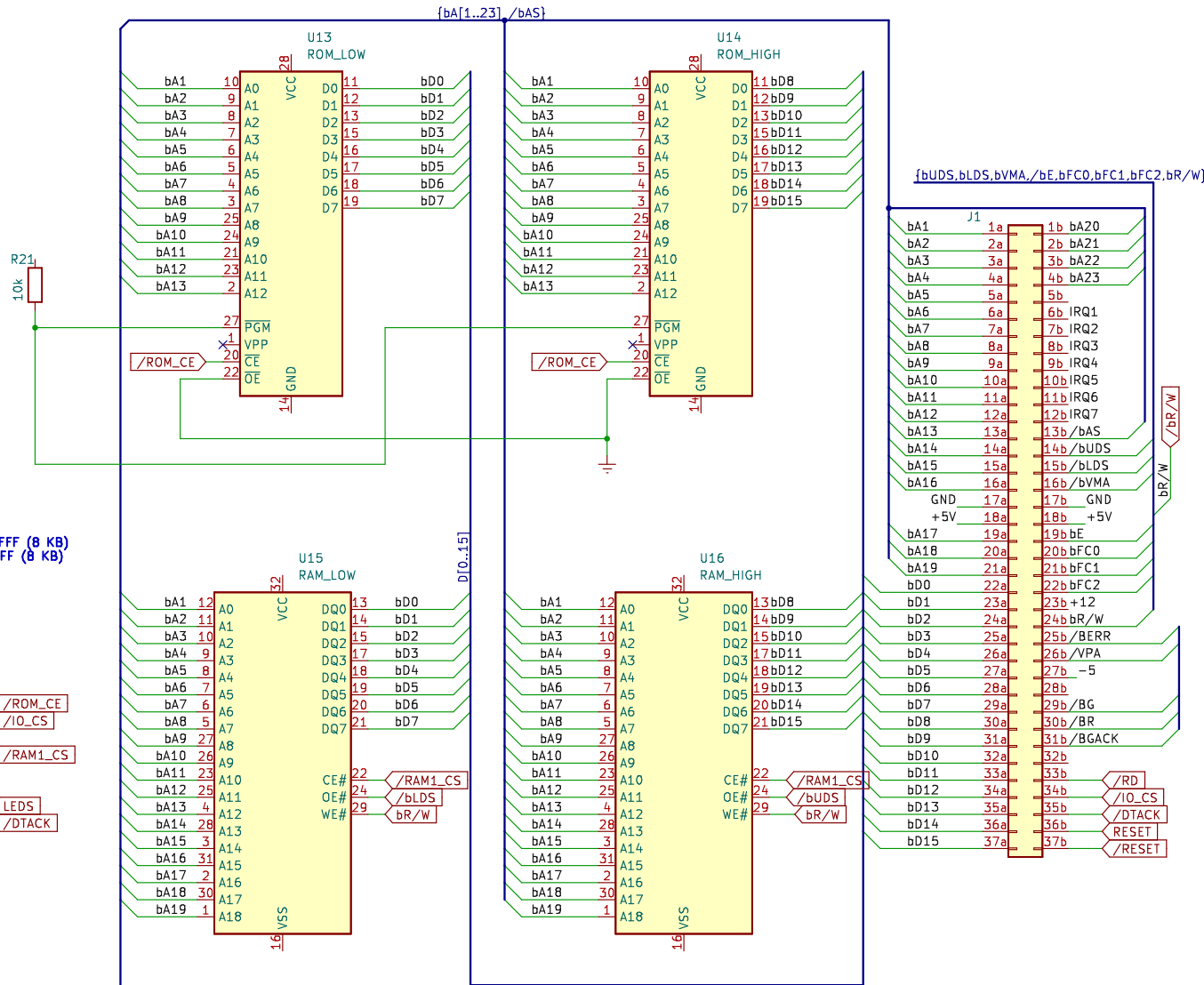
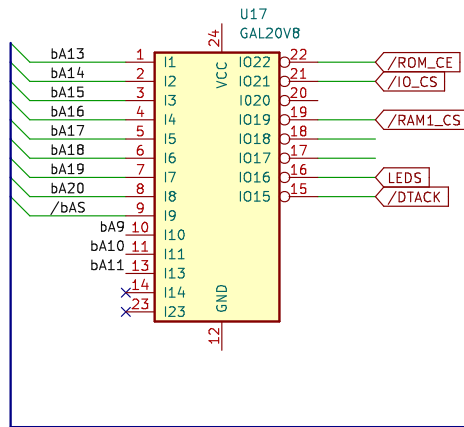
CPU MC68000
ADDRESS BUFFERS
DATA BUFFERS
CONTROL BUFFERS
CLOCK 16MHZ
RESET AND HALT
DTACK GLUE LOGIC

Tie to ground:
DTACK
Tie to VCC:
/BGACK
/BR
/VPA
/BERR
/IPL2
/IPL1
/IPL0
Ignore completely:
FC0
FC1
FC2
/BG
/VMA
E
Tie together
Reset & Halt

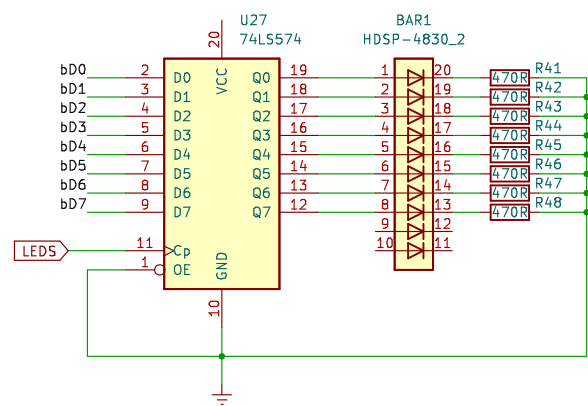
; PINOS DE ENTRADA
A13 PIN 1 : Barramento de endereço
A14 PIN 2
A15 PIN 3
A16 PIN 4
A17 PIN 5
A18 PIN 6
A19 PIN 7
A20 PIN 8
IAS PIN 9 : Address Strobe

; PINOS DE SAÍDA
IROM_CS PIN 19 : ROM (0x00000-0x01FFF)
IIO_CS PIN 18 : I/O (0x02000-0x03FFF)
IRAM4_CS PIN 17 : RAM1 (0x004000-0x07FFFF)
IRAM1_CS PIN 16 : RAM2 (0x080000-0x0FFFFFF)
IRAM2_CS PIN 16 : RAM2 (0x100000-0x17FFFF)
IRAM3_CS PIN 16 : RAM2 (0x180000-0x1FFFFFF)

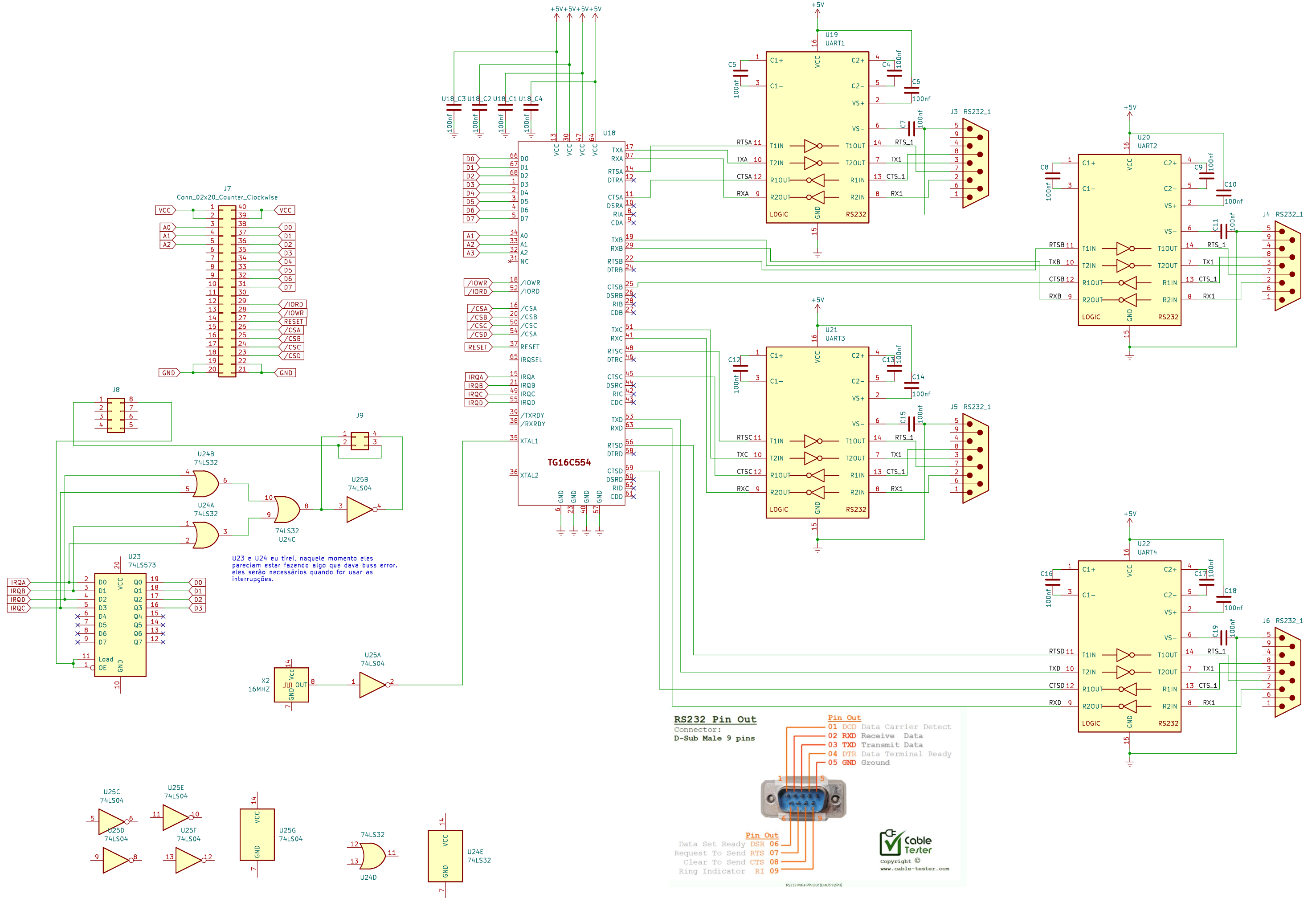
; EQUACÕES
IROM_CS = IAS & IA20 & IA19 & IA18 & IA17 & IA16 & IA15 & IA14 & IA13 : 0x0000-0x1FFF (8 KB)
IIO_CS = IAS & IA20 & IA19 & IA18 & IA17 & IA16 & IA15 & IA14 & A13 : 0x2000-0x3FFF (8 KB)
IRAM4_CS = IAS & IA20 & IA19 : 0x004000-0x07FFFF (20=0 e A19=0, exceto ROM e I/O)
IRAM1_CS = IAS & IA20 & A19 : 0x080000-0x0FFFFFF (20=0 e A19=1)
IRAM2_CS = IAS & A20 & A19 : 0x100000-0x17FFFF (20=1 e A19=0)
IRAM3_CS = IAS & A20 & A19 : 0x180000-0x1FFFFFF (20=1 e A19=1)

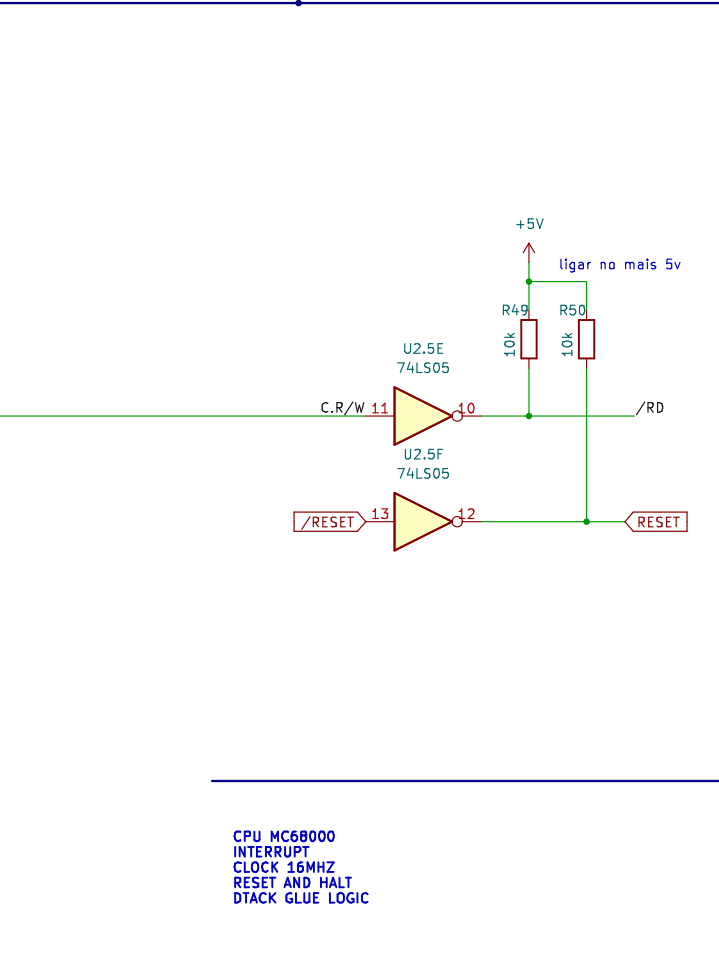
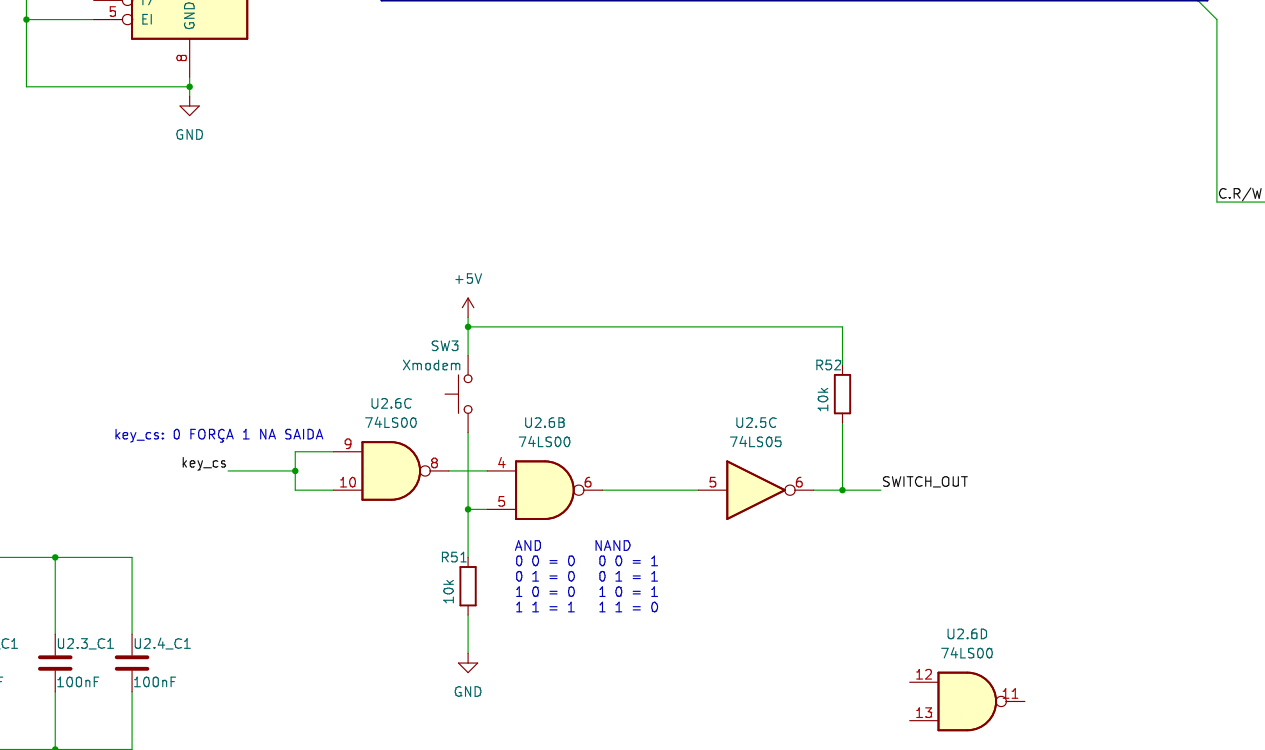
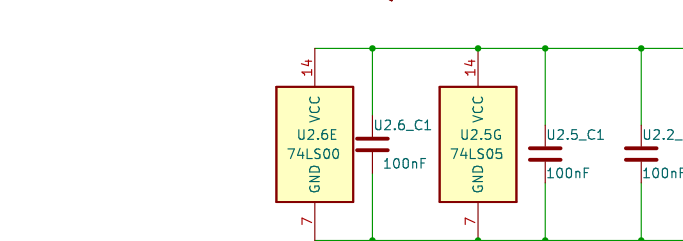


MEMORY CARD
16KB ROM
1MB RAM



Release A: CS unico para as memorias SRAM
Release B: CSH e CSL para as memorias SRAM





CPU MC68000
INTERRUPT
CLOCK 16MHZ
RESET AND HALT
DTACK GLUE LOGIC

UART1 - 0x2000 até 0x20FF = UART_CS1
UART2 - 0x2100 até 0x21FF = UART_CS2
UART3 - 0x2200 até 0x22FF = UART_CS3
UART4 - 0x2300 até 0x23FF = UART_CS4

